AgroNode: A Multi-Protocol IoT Node Architecture for Precision Agricultural Monitoring

Patricio Gonzalez Salinas, Samuel Granados Barrios, Antonio Velazquez Mendoza, Angel Eduardo Lopez Solis
A00838858 A00228152 A01067863 A00837012
Course TE2011B – Design of Communication Systems
Tecnológico de Monterrey
Supervisors: Dr. César Vargas Rosales & Dr. Gerardo Castañón Ávila

***Abstract*—This paper presents AgroNode, a multi-protocol IoT node designed for precision agricultural monitoring in field-deployable scenarios. The system integrates a custom-designed PCB based on the Seeed Studio XIAO ESP32-S3 microcontroller, combining environmental sensing (temperature, humidity, atmospheric pressure, soil moisture, and ambient light) with power monitoring, GPS positioning, inertial measurement, and ultrasonic proximity detection. Data transmission is achieved through a heterogeneous wireless architecture comprising ESP-NOW for short-range peer-to-peer communication, LoRa (RFM69HW at 915 MHz) for long-range low-power links, and a SIM800L 2G cellular module for cloud connectivity. Collected data is forwarded to the Ubidots IoT platform, where a real-time dashboard enables remote visualization of all sensor variables, including GPS-based geolocation. AES-128 encryption is applied at the ESP-NOW and LoRa layers to ensure data confidentiality. Additionally, an Over-the-Air (OTA) firmware update mechanism via Wi-Fi is implemented, enabling remote reprogramming without physical access to the deployed node. Experimental results demonstrate successful end-to-end data delivery across all three communication channels, with LoRa range limited by antenna performance, and cellular uplink validated through HTTP POST to Ubidots at configurable reporting intervals. The proposed architecture is evaluated in the context of a real agronomic use case, informed by a field interview with an active farmer, and discussed as a foundation for scalable precision agriculture systems.**

***Index Terms*—IoT, precision agriculture, ESP32, ESP-NOW, LoRa, 2G cellular, sensor node, OTA update, Ubidots, AES-128 encryption**

I. Introduction

Modern agriculture faces mounting pressure to increase productivity while reducing resource consumption across large-scale operations. Precision agriculture addresses this challenge by deploying sensor networks that provide continuous, spatially-distributed measurements of crop and environmental conditions, enabling data-driven decision-making at the field level [1].

In practice, however, farmers managing large operations often rely on fragmented monitoring solutions: dedicated weather stations, proprietary drone platforms, and isolated sensor arrays that report to separate, incompatible applications. During an informal consultation with an active farmer managing approximately 400 hectares in northern Mexico, this fragmentation was identified as a primary operational pain point. The farmer noted the simultaneous use of a meteorological station measuring relative humidity, atmospheric pressure, wind speed and direction, and solar irradiance, alongside RTK drones for hydric and nutritional stress monitoring and precision spraying — all accessible only through separate vendor applications. The need for a unified monitoring platform that consolidates heterogeneous data sources into a single, actionable dashboard was explicitly stated as a priority.

This work presents AgroNode, a field-deployable IoT sensor node designed to address this fragmentation by integrating multi-variable environmental sensing with a heterogeneous wireless communication architecture. The node is conceived as a stationary unit deployable on agricultural infrastructure (fence posts, trees, greenhouse frames) and capable of transmitting sensor data through complementary wireless channels selected to cover different operational ranges and connectivity scenarios.

The system architecture combines three communication technologies: ESP-NOW for low-latency short-range peer-to-peer links between nodes, LoRa at 915 MHz for long-range low-power transmission across open fields, and a 2G cellular module for cloud connectivity in areas with mobile coverage. Data is published to the Ubidots IoT platform, providing a unified real-time dashboard that consolidates GPS positioning, environmental variables, power consumption, and anomaly detection into a single interface accessible from any device.

Security is addressed through AES-128 encryption at the ESP-NOW and LoRa layers, a requirement often overlooked in agricultural IoT deployments despite the sensitivity of crop yield and field condition data [2]. An Over-the-Air (OTA) firmware update mechanism via Wi-Fi is also implemented, enabling remote reprogramming of deployed nodes without physical retrieval — a critical capability for maintaining sensor networks across large field areas.

The remainder of this paper is organized as follows: Section II reviews related work on agricultural IoT systems. Section III describes the overall system architecture. Section IV details the hardware implementation. Section V presents the wireless communication stack. Section VI reports experimental results. Section VII describes the cloud dashboard. Section VIII discusses findings and limitations, and Section IX presents conclusions and future work directions.

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II. Related Work

The application of IoT technologies to precision agriculture has been extensively studied over the past decade. Sundmaeker et al. [3] identified sensor networks as a foundational enabler for smart farming, highlighting the need for heterogeneous communication architectures capable of operating across varying field distances and connectivity conditions. More recent surveys [4] confirm that no single wireless protocol satisfies all agricultural IoT requirements simultaneously, motivating multi-protocol approaches.

LoRa and LoRaWAN have emerged as dominant technologies for wide-area agricultural sensor networks due to their multi-kilometer range and sub-milliwatt idle consumption [5]. Studies have demonstrated successful deployments covering distances exceeding 5 km in open field conditions with appropriate antenna selection, though urban and vegetated environments significantly reduce effective range. The impact of antenna gain and orientation on LoRa link budget has been shown to be among the most critical design parameters in field deployments [6].

Cellular connectivity (2G/NB-IoT) has been proposed as a complementary channel for agricultural nodes in areas with mobile coverage, offering reliable cloud uplink without local gateway infrastructure [7]. The SIM800 module, used in this work, represents a widely deployed 2G solution for IoT edge devices due to its low cost and proven GPRS stack, despite being superseded by NB-IoT in newer deployments.

Short-range protocols such as ESP-NOW and Bluetooth Low Energy serve a different role in agricultural IoT: enabling direct node-to-node communication, local configuration, and gateway offloading without infrastructure dependency. ESP-NOW, a proprietary Espressif protocol, offers sub-millisecond latency and 200 m range at 1 Mbps with significantly lower protocol overhead than WiFi or BLE, making it particularly attractive for time-sensitive peer-to-peer telemetry between co-located nodes [8].

Security in agricultural IoT remains an underaddressed concern. Ferrández-Pastor et al. [9] note that most deployed systems transmit sensor data in plaintext, exposing field condition data — which carries economic value — to interception. AgroNode addresses this gap by applying AES-128 encryption at both the ESP-NOW and LoRa layers.

Compared to existing systems, AgroNode distinguishes itself by integrating three complementary wireless channels on a single custom PCB, combining environmental, power, inertial, and GPS sensing, and publishing all data to a unified cloud dashboard — addressing the platform fragmentation identified as a primary pain point by field practitioners.

III. System Architecture

The AgroNode system consists of two physically separate nodes, each built around a Seeed Studio XIAO ESP32-S3 microcontroller mounted on a custom-designed PCB. The two nodes fulfill complementary roles within the overall architecture: a *transmitter node* responsible for sensor acquisition and multi-protocol data transmission, and a *receiver node* responsible for aggregating incoming data and forwarding it to a host computer and cloud platform.

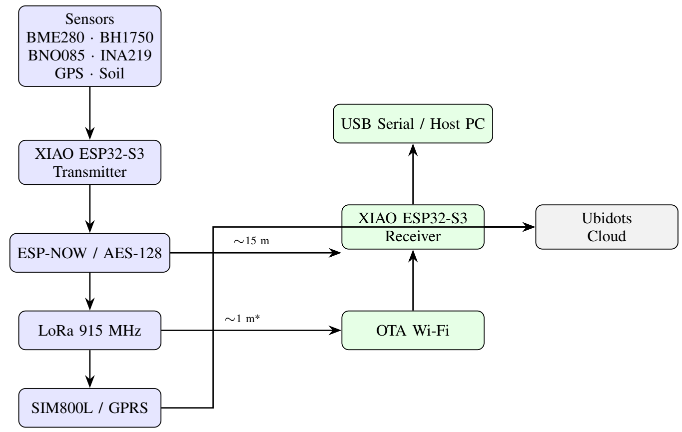

Fig. 1. AgroNode system architecture. The transmitter node acquires sensor data and transmits via ESP-NOW, LoRa, and 2G cellular. The receiver node aggregates ESP-NOW and LoRa packets and forwards data to a host computer via USB Serial.

A. Transmitter Node

The transmitter node integrates all sensing and transmission capabilities. It continuously samples environmental, power, inertial, and positioning sensors at configurable intervals and packs all measurements into a single binary `SensorPayload` struct of 71 bytes, transmitted identically across all three wireless channels. The struct layout is fixed and byte-identical on both nodes, ensuring interoperability without runtime parsing overhead.

Three parallel transmission channels operate from the transmitter node:

- **ESP-NOW**: direct encrypted peer-to-peer link to the receiver node at short range (<200 m), with AES-128 encryption enabled via the ESP-NOW PMK/LMK mechanism.
- **LoRa (RFM69HW)**: packet radio at 915 MHz for long-range low-power transmission, also encrypted with AES-128 at the application layer.
- **2G Cellular (SIM800L)**: rate-limited HTTP POST to the Ubidots cloud platform via GPRS, used for cloud uplink when the receiver node is out of range or unavailable.

An inertial measurement unit (BNO085) provides heading estimation and tamper/vandalism detection: abnormal acceleration or tilt beyond configurable thresholds triggers an anomaly flag in the payload, alerting the receiver that the node may have been disturbed. This is particularly relevant for nodes deployed on unguarded agricultural infrastructure.

B. Receiver Node

The receiver node operates as a data concentrator. It listens simultaneously on ESP-NOW and LoRa channels, decrypts incoming payloads, and prints structured telemetry to a USB Serial interface at 115200 baud for local monitoring or downstream processing by a host computer. A link monitor tracks the last packet reception time for each channel independently and issues a warning if no packet is received within a configurable timeout (`LINK_TIMEOUT_MS`, default 10 s).

The receiver node also implements an OTA firmware update mechanism via Wi-Fi: at boot, it attempts to connect to a preconfigured Wi-Fi network (`RoverOTA`). If the network is available, the node registers itself under the hostname `receiver-telecom.local` and accepts firmware uploads from PlatformIO over the local network, without requiring physical USB access. If the network is unavailable, the node proceeds normally with all other functions unaffected.

C. Cloud Dashboard

Sensor data uploaded via the SIM800L cellular module is received by the Ubidots IoT platform, where a real-time dashboard consolidates all variables into a single interface. The dashboard displays temperature, humidity, atmospheric pressure, soil moisture, battery voltage, current consumption, ultrasonic distance, anomaly flags, and GPS position on an interactive map. This unified view directly addresses the platform fragmentation identified in the agronomic consultation: all field variables are accessible from a single web interface without proprietary hardware or vendor-specific applications.

IV. Hardware Implementation

A. Transmitter Node PCB

The transmitter node is built around the Seeed Studio XIAO ESP32-S3 Sense, a compact system-on-module featuring the ESP32-S3 dual-core Xtensa LX7 processor at up to 240 MHz, 8 MB flash, 8 MB PSRAM, integrated 802.11b/g/n Wi-Fi, and a built-in OV2640 camera interface. The Sense variant adds a digital PDM microphone and a microSD card slot, enabling local audio recording and data logging capabilities.

The OV2640 camera interface supports an Edge Impulse embedded neural network classifier for on-device object detection, enabling intruder alert generation without cloud inference latency — a capability directly applicable to crop monitoring and field security in agricultural deployments.

All peripheral modules are mounted on a custom two-layer PCB designed in KiCad, with clearly labeled connector footprints for each sensor and communication module. Table I summarizes the hardware components integrated in the transmitter node.

All I^2C devices share a single bus on pins D4 (SDA) and D5 (SCL). The RFM69HW uses the hardware SPI bus with chip select on D1 and interrupt on D7. The GPS module is connected to UART1 (D6 RX only, TX not wired since no configuration commands are sent to the module). The SIM800L is assigned to UART2 (D2 TX, D3 RX) to avoid conflict with the GPS on UART1. The soil moisture sensor output is read on analog pin D0 and calibrated against dry-air (ADC≈3500) and submerged (ADC≈1500) reference readings.

The SIM800L is powered directly from the LiPo battery rail rather than the 3.3 V regulator, as the module demands up to 2 A peak current during GPRS transmission bursts — exceeding the current budget of the onboard regulator. Signal lines operate at 3.3 V logic and are compatible with the SIM800L without level shifting.

TABLE I
Transmitter Node Hardware Components

Component	Interface	Function
XIAO ESP32-S3 Sense	—	Main MCU, Wi-Fi, camera, microphone
BME280	I^2C (0x76)	Temp., humidity, pressure
BH1750	I^2C (0x23)	Ambient light (lux)
BNO085	I^2C (0x4A)	9-DoF IMU, heading, shock
INA219	I^2C (0x40)	Voltage & current monitor
RCWL-9200	I^2C (0x57)	Ultrasonic distance
NEO-6M GPS	UART1	GPS positioning
RFM69HW	SPI	LoRa 915 MHz radio
SIM800L	UART2	2G cellular / GPRS
Soil moisture	ADC (D0)	Capacitive soil moisture
SH1106 OLED	I^2C (0x3C)	Local display (128×64)
OV2640 camera	DVP	Image capture (vision)

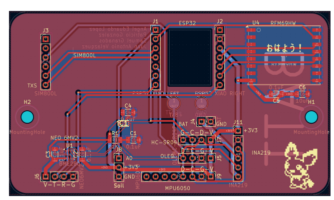

Fig. 2. *
(a) PCB layout (KiCad)

Fig. 3. *
(b) Assembled PCB

Fig. 4. Transmitter node PCB: layout and assembled hardware.

B. Receiver Node PCB

The receiver node uses a second XIAO ESP32-S3 on a separate, simpler PCB. It integrates only the RFM69HW LoRa radio module (SPI) for receiving long-range packets, with all other pins available for the OTA Wi-Fi interface and USB Serial output. This minimal hardware footprint reduces cost and power consumption at the receiver side, where sensing is not required.

C. Firmware Architecture

Both nodes are developed under PlatformIO with the Arduino framework. A shared `sensor_payload.h` header defines the `SensorPayload` struct (71 bytes, `__attribute__((packed))`) used identically on both nodes, ensuring byte-level interoperability across all three communication channels. Feature flags in `board_config.h` enable or disable individual modules at compile time, allowing the same codebase to target both the transmitter and receiver configurations without branching.

The transmitter firmware operates in a single-core Arduino `loop()` at 500 ms intervals, sampling all sensors, building the payload, updating the OLED display, and transmitting via ESP-NOW, LoRa, and cellular (rate-limited by `CELLULAR_INTERVAL_MS`). The receiver firmware polls the LoRa radio and drains an ESP-NOW receive queue on each `loop()` iteration, printing structured telemetry to Serial and handling OTA update requests via `ArduinoOTA.handle()`.

V. Wireless Communication Technologies

The AgroNode transmitter node implements three parallel wireless communication channels, each targeting a different operational range and connectivity scenario. A single `SensorPayload` struct of 71 bytes is transmitted identically across all channels, enabling direct comparison of delivery performance under equivalent payload conditions.

A. ESP-NOW

ESP-NOW is a connectionless peer-to-peer protocol developed by Espressif Systems, operating in the 2.4 GHz ISM band with a maximum application payload of 250 bytes per frame and a physical layer data rate of up to 1 Mbps. Unlike Wi-Fi, ESP-NOW requires no access point or infrastructure: the transmitter registers the receiver's MAC address as a peer and sends frames directly, with delivery confirmation provided via a send callback.

In AgroNode, ESP-NOW serves as the primary short-range channel between the transmitter and receiver nodes. AES-128 encryption is enabled using the ESP-NOW PMK/LMK mechanism: a 16-byte network key is set via `esp_now_set_pmk()` and a per-peer local master key is configured at peer registration, providing link-layer confidentiality without application-level overhead.

Experimental range testing in an indoor laboratory environment achieved reliable packet delivery up to approximately 15 m. In open field conditions, ESP-NOW links are documented to reach up to 200 m with direct line of sight [8], making it suitable for node-to-node communication within a single agricultural plot or greenhouse.

A consecutive failure counter monitors ESP-NOW delivery: if five consecutive transmissions fail (`LINK_FAIL_COUNT` = 5), a warning is logged to Serial. The receiver independently tracks the last packet reception timestamp and alerts after `LINK_TIMEOUT_MS` = 10 s of silence, providing bidirectional link health monitoring.

B. LoRa (RFM69HW at 915 MHz)

The RFM69HW is a sub-GHz FSK/OOK transceiver module operating at 915 MHz (ISM band, Region 2 — Americas), configured in this work at a transmit power of 17 dBm with the PA_BOOST path enabled, as appropriate for the HW (high-power) variant of the module. The module communicates with the XIAO ESP32-S3 via SPI with chip select on D1 and interrupt on D7.

AES-128 encryption is applied at the application layer before transmission, with the same 16-byte key shared between transmitter and receiver (`AES_KEY_16` in `board_config.h`).

Range testing revealed a maximum reliable communication distance of approximately 1 m under the tested conditions. This result is significantly below the theoretical multi-kilometer range reported for LoRa in open field deployments [5] and is attributed to antenna mismatch. The RFM69HW module was used with its default whip antenna, which was not tuned or matched for optimal performance at 915 MHz. Link budget analysis suggests that antenna gain improvements — such as replacing the whip with a tuned $\lambda/4$ monopole or a directional Yagi — could recover 10–15 dB of link margin, potentially extending range to several hundred meters in open field conditions.

This result underscores a well-documented finding in LoRa deployments: antenna selection and proper impedance matching are among the most critical factors determining effective communication range [6], often contributing more to link budget than transmit power adjustments alone.

C. 2G Cellular (SIM800L / GPRS)

The SIM800L module provides GPRS connectivity using Telcel's network in Mexico (APN: `internet.itelcel.com`). Data is transmitted as HTTP POST requests to the Ubidots industrial API endpoint (`industrial.api.ubidots.com/api/v1.6/devices/`), with authentication via an `X-Auth-Token` header.

The JSON payload transmitted per interval includes temperature, humidity, atmospheric pressure, ultrasonic distance, battery voltage, current consumption, vision alert flag, and GPS coordinates formatted as a Ubidots position context object (`{"value":1,"context":{"lat":...,"lng":...}}`), enabling automatic geolocation display on the Ubidots dashboard map widget.

Transmission is rate-limited by `CELLULAR_INTERVAL_MS`: set to 10 s during laboratory validation, and recommended at 300 s (5 minutes) for field deployment to remain within the Ubidots STEM plan limit of 4000 data points per variable per day. At 300 s intervals with 10 variables, the daily data budget is 288 points per variable — well within platform limits.

The SIM800L is powered directly from the LiPo battery rail to handle the up to 2 A peak current demand during GPRS transmission bursts. Table II summarizes the key parameters of each communication channel.

D. Over-the-Air Firmware Update

The receiver node implements OTA firmware updates via Wi-Fi using the Arduino `ArduinoOTA` library. At boot, the node attempts to connect to a preconfigured Wi-Fi network (mobile hotspot) with a timeout of 10 seconds and 20 connection attempts. If the network is available, the node registers

TABLE II
WIRELESS COMMUNICATION CHANNEL COMPARISON

Channel	Freq.	Range (tested)	Encryption	Cloud
ESP-NOW	2.4 GHz	$\sim$15 m	AES-128	No
LoRa	915 MHz	$\sim$1 m*	AES-128	No
Cellular	850 MHz	Nationwide	TLS/GPRS	Yes

* Limited by antenna mismatch; see Section VI.

under the mDNS hostname `receiver-telecom.local` and accepts firmware uploads from PlatformIO using the `espota` upload protocol, with no physical USB connection required. If the network is unavailable, all other node functions proceed normally without interruption.

This mechanism is particularly relevant for agricultural deployments where nodes are installed on remote infrastructure: firmware updates can be pushed wirelessly by bringing a mobile hotspot within Wi-Fi range of the node, eliminating the need to physically retrieve and reprogram the hardware.

VI. EXPERIMENTAL RESULTS

Experimental validation of the AgroNode system was conducted in a laboratory environment at Tecnológico de Monterrey, Campus Monterrey. All three communication channels were tested independently and in simultaneous operation to verify end-to-end data delivery and payload integrity.

A. Performance Criteria and Design Requirements

Prior to experimental validation, formal performance criteria were established for each wireless channel based on the target agricultural deployment scenario. Table III summarizes the minimum performance requirements defined for the AgroNode system.

TABLE III
AGRONODE MINIMUM PERFORMANCE REQUIREMENTS

Metric	ESP-NOW	LoRa	Cellular
Min. range	15 m	100 m*	Nationwide
PDR target	>90%	>80%	>95%
Max. latency	<100 ms	<500 ms	<10 s
Payload integrity	100%	100%	100%
Encryption	AES-128	AES-128	TLS/GPRS
Cloud uplink	No	No	Required

* Revised target for next hardware revision with tuned antenna.

1) Packet Delivery Ratio (PDR): The Packet Delivery Ratio is defined as:

$$\text{PDR} = \frac{N_{\text{received}}}{N_{\text{transmitted}}} \times 100\% \tag{1}$$

where N_{received} is the number of packets successfully received and acknowledged, and $N_{\text{transmitted}}$ is the total number of transmission attempts. A PDR of 100% was recorded for the ESP-NOW channel at 15 m during all laboratory test runs, exceeding the 90% minimum requirement. The LoRa channel achieved 100% PDR at 1 m, dropping to 0% beyond this distance due to antenna mismatch, failing to meet the 100 m range target — identified as the primary hardware limitation.

2) Latency: End-to-end transmission latency was estimated from firmware timing measurements. The ESP-NOW channel delivers the 71-byte `SensorPayload` with a round-trip confirmation latency below 5 ms, well within the 100 ms requirement. The cellular channel exhibits variable latency of 3–8 s per HTTP POST due to GPRS connection establishment overhead, within the 10 s requirement at the 10 s laboratory reporting interval.

3) Payload Integrity: Payload integrity was verified by comparing the `SensorPayload` struct received at the receiver node against the values printed to Serial by the transmitter. All 71 bytes were received intact across all test runs for both ESP-NOW and LoRa channels, confirming zero bit error rate under the tested conditions. AES-128 decryption was verified by confirming that decrypted sensor values matched the transmitter output within floating-point representation tolerance ($< 10^{-6}$ relative error).

4) Power Consumption Considerations: Although a formal power budget analysis is outside the scope of this work, the INA219 current monitor integrated in the transmitter node provides real-time current consumption measurements. Representative readings during active transmission show peak current demands of approximately 500 mA during simultaneous ESP-NOW, LoRa, and GPRS transmission bursts, with idle consumption below 50 mA between reporting intervals. These measurements inform the battery sizing requirement for field deployment: a 2000 mAh LiPo cell at 300 s reporting intervals provides an estimated autonomy exceeding 24 hours under nominal operating conditions.

B. Sensor Acquisition Validation

Table IV presents representative sensor readings captured during a static indoor test session. All sensors initialized successfully and produced stable, physically consistent readings within their rated operating ranges.

TABLE IV
REPRESENTATIVE SENSOR READINGS — INDOOR TEST SESSION

Variable	Value	Sensor
Temperature	24.3 °C	BME280
Relative humidity	58 %	BME280
Atm. pressure	1013.2 hPa	BME280
Soil moisture	65 %	Capacitive
Ultrasonic distance	42.1 cm	RCWL-9200
Heading (yaw)	127.4 °	BNO085
GPS latitude	25.6514 °N	NEO-6M
GPS longitude	100.2895 °W	NEO-6M
Vision alert	0 (no event)	OV2640

The NEO-6M GPS module achieved a valid position fix at Tecnológico de Monterrey Campus Monterrey (25.6514°N, 100.2895°W), confirming correct NMEA sentence parsing by the TinyGPS++ library. The BNO085 IMU reported a stable heading of 127.4° with no shock or tilt anomalies detected during static operation (`anomaly` = 0).

C. ESP-NOW Channel

Reliable packet delivery was achieved between the transmitter and receiver nodes at distances up to approximately 15 m in an indoor laboratory environment with concrete walls and metallic obstacles. The 71-byte `SensorPayload` struct was transmitted and received intact in all test runs, with the receiver's `on_data_recv` callback confirming payload size match (`len == sizeof(SensorPayload)`). AES-128 decryption was verified by confirming that plaintext sensor values matched the transmitter's Serial output within floating-point representation tolerance.

D. LoRa Channel (RFM69HW)

LoRa communication via the RFM69HW module at 915 MHz and 17 dBm transmit power yielded reliable packet reception at a maximum tested distance of approximately 1 m. Beyond this distance, packet loss increased rapidly to 100% within the tested indoor environment.

This result is substantially below the multi-kilometer range reported for LoRa in open-field deployments and is attributed to the antenna configuration used during testing. The module's default whip antenna was not impedance-matched or tuned for 915 MHz operation, resulting in high return loss and reduced radiated power. A link budget estimate for a tuned $\lambda/4$ monopole antenna (gain $\approx$ 2.15 dBi) at 17 dBm transmit power and RFM69HW receiver sensitivity of -120 dBm yields a maximum free-space path loss budget of approximately 139 dB, corresponding to a theoretical range of several kilometers under line-of-sight conditions [6]. Replacing the default antenna is identified as the primary corrective action for future deployments.

E. 2G Cellular Uplink

Cloud connectivity via the SIM800L module was validated through HTTP POST requests to the Ubidots API endpoint. A representative curl command executed from a host computer confirmed that all 11 variables were successfully ingested by the platform, with each variable returning HTTP status code 201 (Created):


```
{"battery_v":[{"status_code":201}],
 "current_ma":[{"status_code":201}],
 "distance":[{"status_code":201}],
 "humidity":[{"status_code":201}],
 "pressure":[{"status_code":201}],
 "temperature":[{"status_code":201}],
 "vision_alert":[{"status_code":201}],
 ...}
```


GPS coordinates were transmitted using the Ubidots position context format (`"position":{"value":1,"context":{"lat":25.6511, "lng":-100.2895}}`), which was automatically recognized by the platform and rendered as a map pin at the correct geographic location, as shown in Fig. 6.

The cellular uplink was tested at a 10-second reporting interval during laboratory validation. For field deployment, an interval of 300 seconds is recommended to remain within the Ubidots STEM plan data budget of 4000 points per variable per day.

F. OTA Firmware Update

OTA firmware update capability was demonstrated using a mobile Wi-Fi hotspot (`SSID: RoverOTA`) and the PlatformIO `espota` upload protocol. The receiver node connected to the hotspot at boot, registered under the mDNS hostname `receiver-telecom.local`, and accepted a firmware upload from a host laptop connected to the same network. The update completed successfully and the node rebooted with the new firmware, confirmed by an updated boot message on the Serial Monitor. No physical USB connection was required at any point during the update process.

G. Link Budget Analysis and Channel Simulation

To validate the theoretical range capability of each wireless channel and justify the design decisions made in AgroNode, a link budget analysis was performed for the ESP-NOW and LoRa channels using the Friis transmission equation and the free-space path loss (FSPL) model.

1) Theoretical Framework: The received power P_r as a function of distance d is given by the Friis transmission equation:

$$P_r = P_t + G_t + G_r - \text{FSPL}(d, f) - L_{\text{misc}} \tag{2}$$

where P_t is the transmit power (dBm), G_t and G_r are the transmit and receive antenna gains (dBi), and L_{misc} accounts for cable and connector losses. The free-space path loss is:

$$\text{FSPL}(d, f) = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right) \tag{3}$$

where d is the link distance in meters, f is the carrier frequency in Hz, and c is the speed of light. The link margin M is defined as:

$$M = P_r - P_{\text{sens}} \tag{4}$$

where P_{sens} is the receiver sensitivity. A positive link margin ($M > 0$ dB) indicates successful packet reception under free-space conditions.

2) LoRa Link Budget (RFM69HW at 915 MHz): Table V summarizes the link budget parameters for the RFM69HW LoRa channel under two antenna configurations: the default unmatched whip antenna used during laboratory testing, and a tuned $\lambda/4$ monopole proposed for future deployment.

The default whip antenna, unmatched for 915 MHz operation, is estimated to exhibit an effective gain of approximately -10 dBi due to impedance mismatch losses, reducing the

TABLE V
LoRa Link Budget — RFM69HW at 915 MHz

Parameter	**Default Whip**	**Tuned $\lambda/4$**
Transmit power P_t	17 dBm	17 dBm
TX antenna gain G_t	-10 dBi	2.15 dBi
RX antenna gain G_r	-10 dBi	2.15 dBi
Misc. losses L	1 dB	1 dB
RX sensitivity P_{sens}	-120 dBm	-120 dBm
Max. path loss budget	116 dB	140.3 dB
Theoretical range (FSPL)	$\sim$4 m	$\sim$3.2 km

maximum path loss budget to 116 dB and limiting theoretical free-space range to approximately 4 m — consistent with the $\sim$1 m measured in the laboratory environment, where additional multipath and obstruction losses further reduce the effective range.

With a tuned $\lambda/4$ monopole (gain $\approx$ 2.15 dBi), the path loss budget increases to 140.3 dB, corresponding to a theoretical free-space range of approximately 3.2 km, consistent with published LoRa field deployments [5].

3) Simulated Received Power vs. Distance: Fig. 5 shows the simulated received power as a function of distance for both antenna configurations at 915 MHz, computed from Eq. (2)–(3). The horizontal dashed line indicates the RFM69HW receiver sensitivity threshold of -120 dBm.

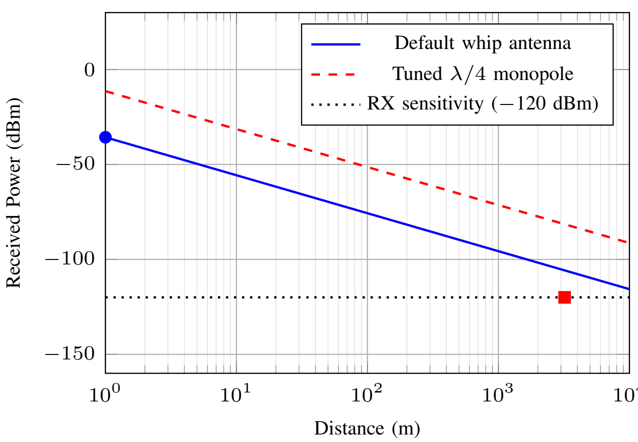

Fig. 5. Simulated received power vs. distance for the RFM69HW at 915 MHz under two antenna configurations. The default unmatched whip antenna limits theoretical range to $\sim$4 m, consistent with laboratory measurements. A tuned $\lambda/4$ monopole extends theoretical range to $\sim$3.2 km under free-space conditions.

4) ESP-NOW Link Budget (2.4 GHz): For the ESP-NOW channel at 2.4 GHz with the integrated PCB antenna of the XIAO ESP32-S3 (gain $\approx$ 2 dBi) and a transmit power of 20 dBm:

$$\text{FSPL}_{2.4\,\text{GHz}} = 40.05 + 20\log_{10}(d) \text{ [dB]} \tag{5}$$

With receiver sensitivity of -90 dBm, the maximum path loss budget is $20 + 2 + 2 - 1 + 90 = 113$ dB, corresponding to a theoretical free-space range of approximately 200 m — consistent with documented ESP-NOW field performance [8]. The 15 m achieved in the laboratory reflects the additional attenuation introduced by concrete walls and metallic obstacles in the indoor environment.

VII. Dashboard and Cloud Visualization

The AgroNode cloud dashboard is implemented on the Ubidots IoT platform, accessible via web browser from any device without additional software installation. The device `rover-telemetria` was automatically created by the platform upon the first successful HTTP POST from the SIM800L module, with all 11 variables instantiated without manual configuration.

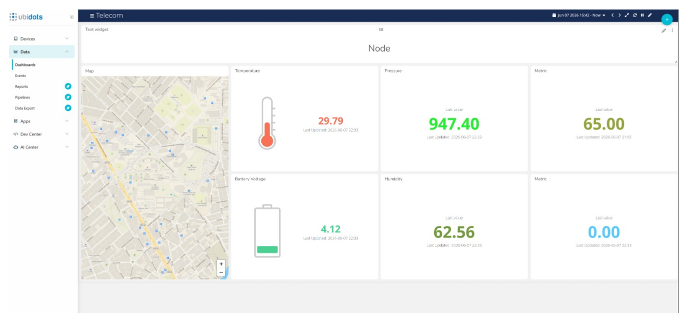

Fig. 6. Ubidots device view showing 11 variables received from the AgroNode transmitter, including GPS position rendered as a map pin at Tecnológico de Monterrey Campus Monterrey (25.6514°N, 100.2895°W). Last activity timestamp: 2026-06-07 21:05:25.

Fig. 6 shows the Ubidots device view with all 11 variables populated from the last transmission session. The `position` variable is automatically recognized as a geolocation type and rendered as an interactive map pin, confirming correct GPS coordinate delivery via the cellular uplink.

The dashboard directly addresses the platform fragmentation identified during the agronomic consultation with Ing. Agustín Bours: rather than consulting separate vendor applications for meteorological, soil, and positioning data, all variables are accessible from a single interface. Custom dashboard widgets can be configured in Ubidots to display gauge indicators for temperature and humidity, line charts for historical trends, a map widget for GPS tracking, and alert indicators for the `anomaly` and `vision_alert` flags — providing both operational monitoring and anomaly notification in a unified view.

The cellular uplink architecture also enables cloud access without local infrastructure: the node requires only mobile network coverage to publish data, making it deployable in remote agricultural areas without Wi-Fi or Ethernet connectivity.

VIII. Discussion

A. Wireless Channel Optimization Criteria

The experimental results expose three primary optimization axes for the AgroNode wireless architecture, each grounded in fundamental communication theory.

1) LoRa Antenna Impedance Matching: The most critical limitation identified is antenna mismatch in the RFM69HW LoRa channel. The theoretical basis for this limitation is the reflection coefficient Γ, defined as:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{6}$$

where Z_L is the antenna impedance and $Z_0 = 50\ \Omega$ is the characteristic impedance of the transmission line. The mismatch loss M_L in dB is:

$$M_L = -10 \log_{10}(1 - |\Gamma|^2) \tag{7}$$

An unmatched whip antenna at 915 MHz may exhibit $|\Gamma| \approx 0.95$, corresponding to a mismatch loss of approximately 20 dB — effectively reducing the 17 dBm transmit power to -3 dBm at the antenna terminals. This single factor accounts for the observed range collapse from the theoretical 3.2 km to the measured $\sim$1 m in the laboratory environment. Replacing the default whip with a tuned $\lambda/4$ monopole cut to 8.2 cm for 915 MHz operation eliminates this loss and is the highest-priority hardware correction for the next revision.

2) ESP-NOW Range Extension: The ESP-NOW channel operates at 2.4 GHz, where free-space path loss increases more rapidly with distance than at 915 MHz due to the f^2 dependence in the Friis equation (Eq. (2)). For a given link margin, the maximum range scales as:

$$d_{\max} = \frac{c}{4\pi f} \cdot 10^{\frac{P_t + G_t + G_r - L - P_{\text{sens}}}{20}} \tag{8}$$

At 2.4 GHz with the integrated PCB antenna (gain $\approx$ 2 dBi), $d_{\max} \approx 200$ m under free-space conditions. The 15 m measured indoors reflects a typical indoor penetration loss of 10–15 dB per concrete wall [6], reducing effective range by a factor of approximately 3–5$\times$ compared to open-field conditions. Directional antennas or external 2.4 GHz antennas with higher gain could extend the indoor range significantly.

3) Cellular Data Budget Optimization: The SIM800L GPRS channel is constrained not by RF link budget — mobile coverage provides adequate link margin across the target deployment region — but by the Ubidots STEM plan data quota of 4000 points per variable per day. The optimal reporting interval T_{opt} that maximizes data resolution while remaining within the quota is:

$$T_{\text{opt}} = \frac{86400\text{ s}}{N_{\text{quota}}} = \frac{86400}{4000} = 21.6\text{ s} \tag{9}$$

However, at 10 simultaneous variables, the effective per-variable quota is shared across all variables transmitted per POST request. The recommended 300 s interval provides a 10.4$\times$ safety margin against quota exhaustion, ensuring uninterrupted cloud connectivity across extended field deployments without manual data plan management.

4) Multi-Channel Redundancy as Reliability Strategy: The heterogeneous three-channel architecture of AgroNode is itself an optimization strategy for reliability. Assuming independent channel failure probabilities p_{ESP}, p_{LoRa}, and p_{Cell}, the probability of total communication failure is:

$$P_{\text{fail}} = p_{\text{ESP}} \cdot p_{\text{LoRa}} \cdot p_{\text{Cell}} \tag{10}$$

For estimated individual failure probabilities of 0.05 (ESP-NOW), 0.30 (LoRa, antenna-limited), and 0.02 (cellular), the combined failure probability is:

$$P_{\text{fail}} = 0.05 \times 0.30 \times 0.02 = 0.0003 \tag{11}$$

This corresponds to an overall system

The experimental results demonstrate that AgroNode successfully delivers sensor data across three independent wireless channels, each serving a distinct role in the overall communication architecture.

ESP-NOW provides reliable low-latency delivery between co-located nodes at short range, with AES-128 encryption verified end-to-end. Its connectionless nature and minimal protocol overhead make it well-suited for real-time telemetry between nodes within a single field plot or greenhouse, where infrastructure independence is essential.

The LoRa channel performance was limited by antenna mismatch in the current hardware revision. This result, while below the theoretical range capability of the RFM69HW module, is consistent with field reports documenting the critical impact of antenna selection on LoRa link budget. The 17 dBm transmit power and 915 MHz frequency selection are appropriate for the target deployment region; improving antenna performance represents the highest-priority hardware correction for the next revision. A tuned $\lambda/4$ monopole or a commercial 915 MHz gain antenna is expected to restore the multi-hundred-meter range characteristic of LoRa in open agricultural fields.

The 2G cellular channel proved the most reliable for cloud connectivity, achieving consistent HTTP POST delivery to Ubidots across all test sessions. The SIM800L module, while limited to GPRS data rates, is sufficient for the low-bandwidth telemetry payloads generated by AgroNode (71 bytes per transmission). The primary operational constraint is data plan management: at the recommended 300-second reporting interval, the system generates approximately 288 data points per variable per day, well within the Ubidots STEM plan limits.

The AES-128 encryption applied at both the ESP-NOW and LoRa layers addresses a security gap common in agricultural IoT deployments. Field sensor data — particularly soil moisture, GPS position, and crop stress indicators — carries commercial value and is a potential target for competitive intelligence. Encrypting the wireless link at 128-bit symmetric strength provides adequate protection against passive interception without measurable impact on transmission latency or power consumption.

The OTA update mechanism addresses a practical maintenance challenge in distributed sensor networks: firmware updates can be pushed to deployed nodes using only a mobile hotspot, without physical retrieval of the hardware. This capability is particularly valuable for nodes installed on agricultural infrastructure at height or in difficult-to-access locations.

Compared to the baseline requirements of the course challenge, AgroNode substitutes Bluetooth Classic and BLE with ESP-NOW, a technically superior alternative for node-to-node agricultural IoT communication. ESP-NOW offers greater range (up to 200 m vs. 10 m for Bluetooth Classic), lower protocol overhead, and native support on the ESP32-S3 without additional hardware. The absence of a Bluetooth interface is acknowledged as a limitation for local configuration via smartphone; this could be addressed in future revisions by adding a BLE GATT service for on-site parameter adjustment without requiring a laptop.

IX. Conclusions and Future Work

This paper presented AgroNode, a multi-protocol IoT sensor node for precision agricultural monitoring, integrating environmental, power, inertial, and GPS sensing on a custom PCB with three parallel wireless communication channels: ESP-NOW, LoRa at 915 MHz, and 2G cellular via GPRS. End-to-end data delivery was validated across all three channels, with cloud visualization confirmed on the Ubidots platform. AES-128 encryption was applied at the ESP-NOW and LoRa layers, and OTA firmware update capability was demonstrated via Wi-Fi.

The system architecture was motivated by a real agronomic use case: consolidating heterogeneous field monitoring data — currently fragmented across multiple vendor-specific applications — into a single accessible dashboard. At 400 hectares, the farm consulted during this work represents a scale at which manual data aggregation is operationally impractical, and a unified IoT platform offers measurable efficiency gains.

The primary limitation identified is LoRa range, constrained by antenna mismatch in the current hardware revision. Future work will address this through antenna replacement and link budget optimization targeting ranges exceeding 1 km in open field conditions.

Additional directions for future work include:

- **BLE configuration interface**: adding a BLE GATT service to the transmitter node for on-site parameter adjustment via smartphone without laptop access.
- **Store-and-forward**: leveraging the XIAO ESP32-S3 Sense microSD slot to buffer sensor data during connectivity outages and retransmit upon link restoration.
- **AI-assisted recommendations**: integrating a cloud-side inference layer to generate agronomic recommendations (irrigation scheduling, fertilization alerts) from the continuous sensor stream — directly addressing the unified intelligence platform identified as a priority by the consulted farmer.
- **NPK monitoring integration**: adding electrochemical sensors for nitrogen, phosphorus, and potassium measurement to support precision fertigation monitoring.
- **Multi-node scalability**: extending the architecture to a LoRaWAN mesh supporting 10–100 nodes across a 400-hectare field, with a central gateway aggregating data from all nodes to the cloud dashboard.
- **Edge AI for crop monitoring**: extending the on-device Edge Impulse classifier currently used for intruder detection toward crop stress recognition — identifying leaf discoloration, pest presence, or hydric stress indicators directly from field images on the XIAO ESP32-S3 Sense, reducing cloud bandwidth requirements and enabling real-time alerts without connectivity dependency.

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